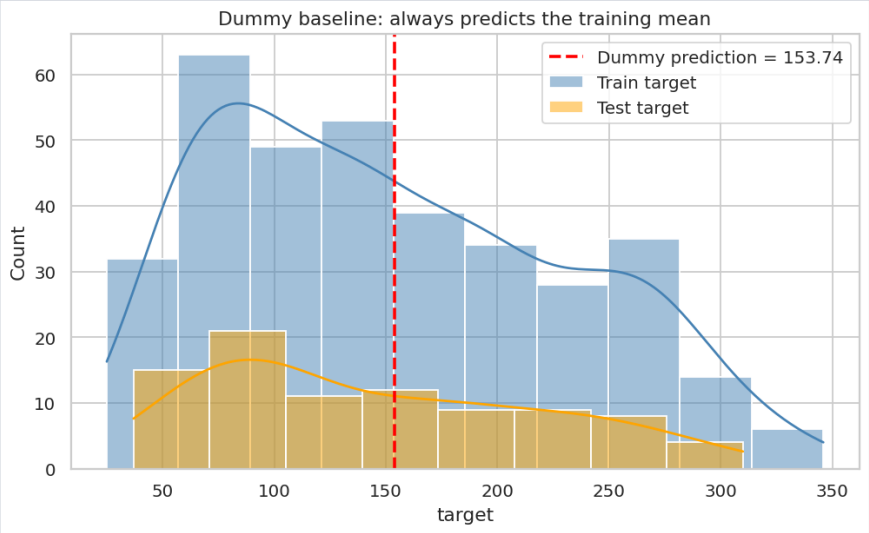
1) Dummy mean baseline

Target distribution vs constant prediction

RMSE	73.22
MAE	64.01
R^2	-0.012
Fit time (s)	—

Always predicts the training mean (153.74). The histogram shows train and test targets are right-skewed, so a constant guess misses both tails. R^2 is about zero by construction, which makes this the floor every other model must beat.

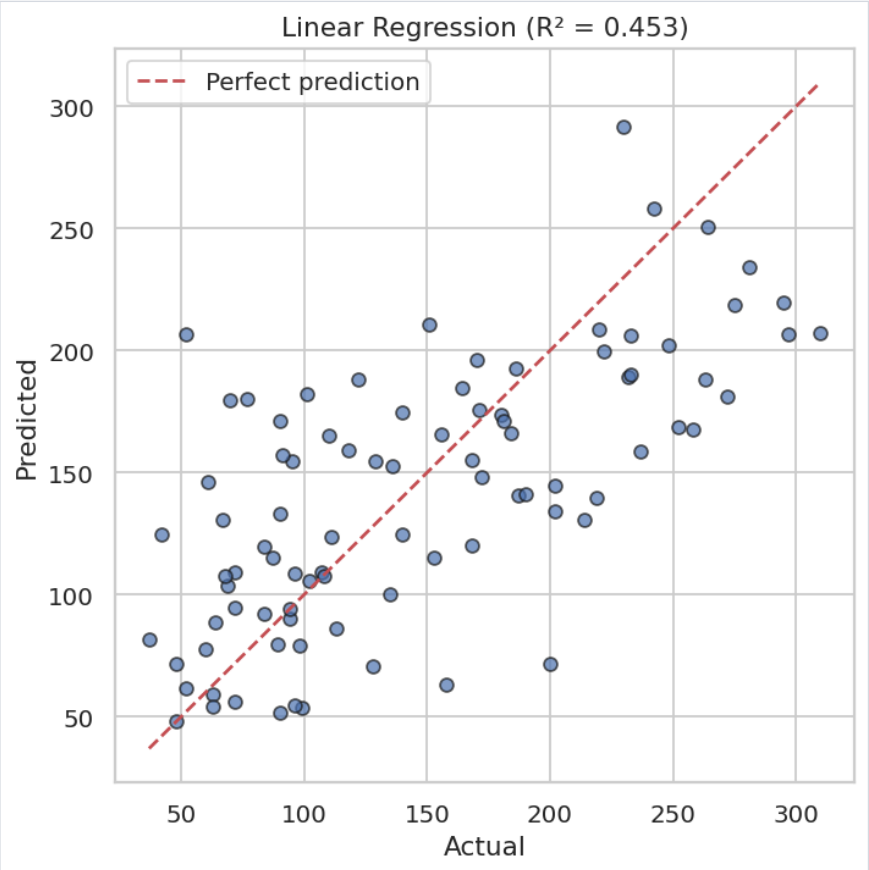


2) Linear Regression

Actual vs predicted

RMSE	53.85
MAE	42.79
R^2	0.453
Fit time (s)	0.123

Points follow the diagonal loosely. Low targets tend to be over-predicted and high targets under-predicted, a typical regression-to-the-mean pattern. This is the reference linear fit that the regularised variants are compared with.

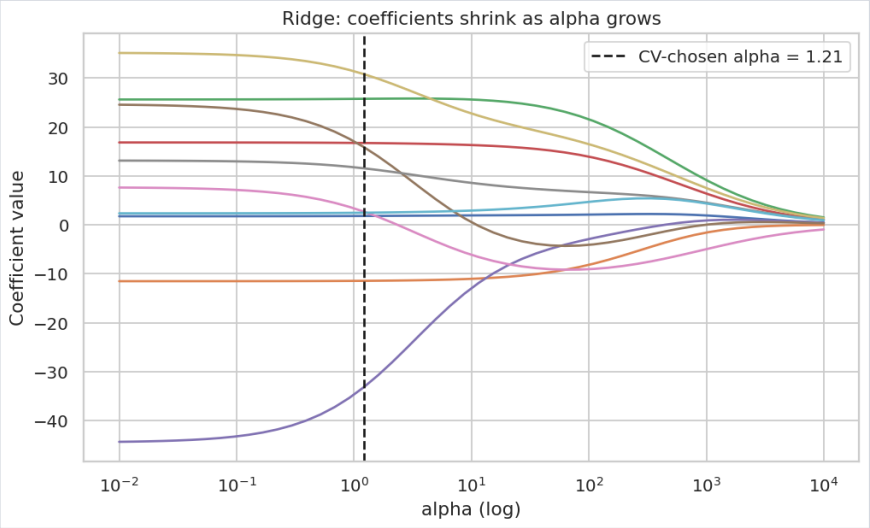


3) Ridge

Coefficient shrinkage path

RMSE	53.77
MAE	42.81
R^2	0.454
Fit time (s)	0.007

Each line is one feature coefficient as alpha grows (log scale). All coefficients shrink smoothly toward zero but none reach it. The dashed line marks the CV-chosen alpha of 1.21, where the fit is almost identical to OLS.

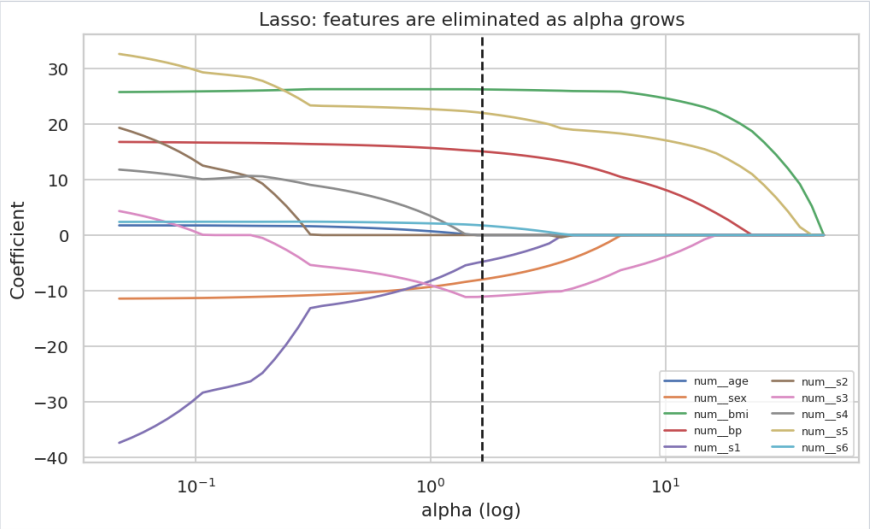


4) Lasso

Coefficient path (features dropping to zero)

RMSE	52.92
MAE	42.79
R^2	0.471
Fit time (s)	0.047

Coefficients hit exactly zero one by one as alpha increases, performing feature selection. BMI and s5 survive the longest. At the CV-chosen alpha (dashed line) several weaker features are already removed, and the sparser model beats plain OLS.

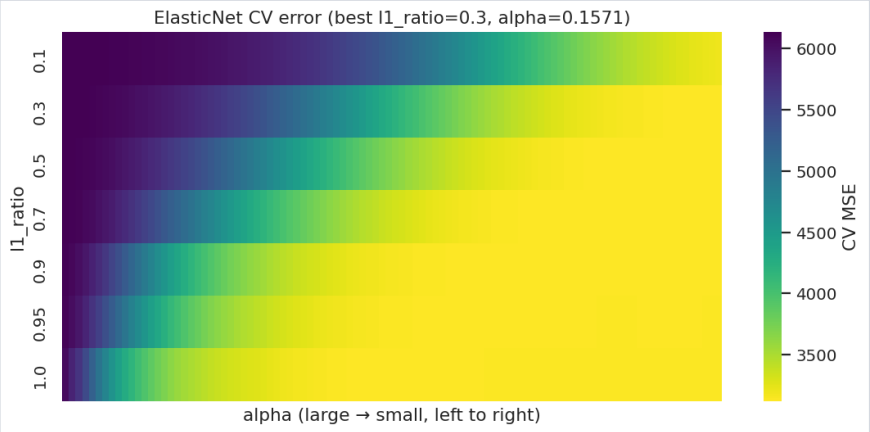


5) ElasticNet

CV error heatmap (alpha x l1_ratio)

RMSE	53.45
MAE	42.98
R^2	0.461
Fit time (s)	0.454

Cross-validated MSE over the alpha and l1_ratio grid. Error is lowest (yellow) at small alpha and rises sharply (purple) when alpha is large. The best combination is l1_ratio = 0.3 with alpha = 0.1571. The 0.454 s fit time includes this CV search.

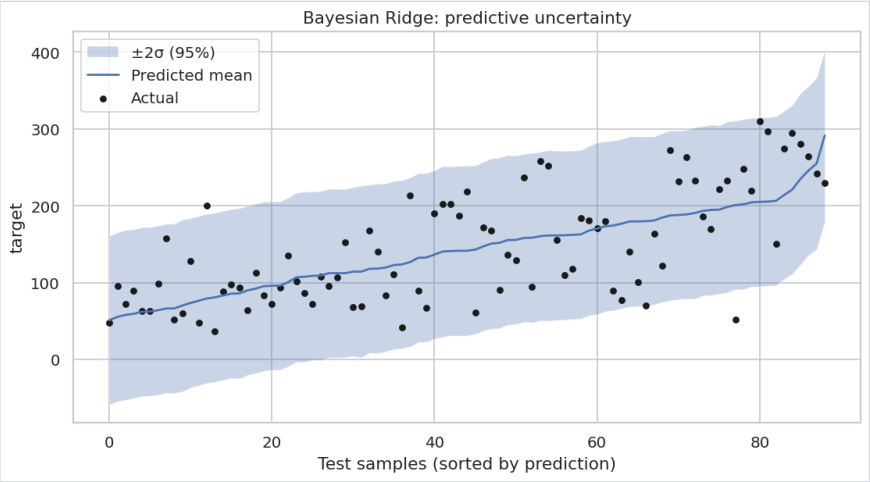


6) Bayesian Ridge

Predictions with uncertainty band

RMSE	53.59
MAE	42.87
R^2	0.458
Fit time (s)	0.005

Test samples sorted by predicted value, with the predicted mean and a 95% (+/-2 sigma) band. The band is wide (roughly +/-100 units) and most actual values fall inside it, so the model is honest about how noisy this target is.

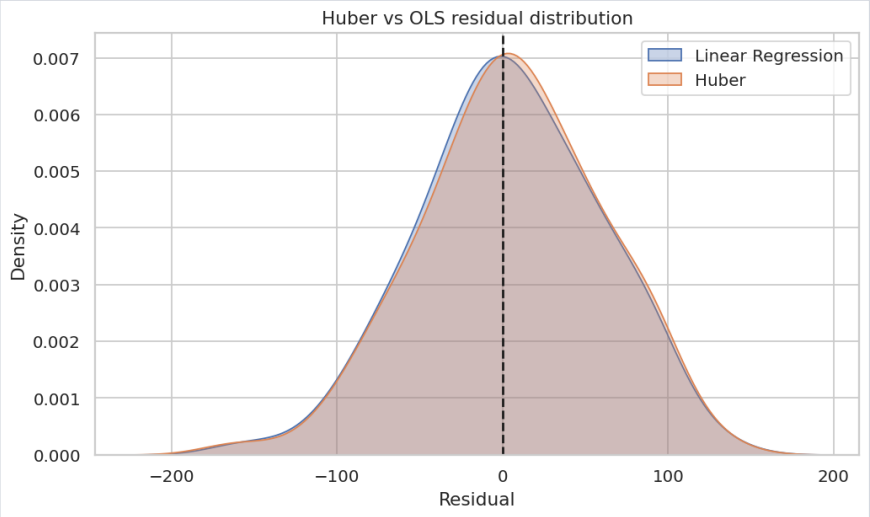


7) Huber Regressor

Outlier handling (residuals vs OLS)

RMSE	54.14
MAE	42.88
R^2	0.447
Fit time (s)	0.007

Residual densities for Huber and OLS almost overlap. The data contain few extreme outliers, so the robust loss changes very little, and R^2 is marginally lower (0.447 vs 0.453).

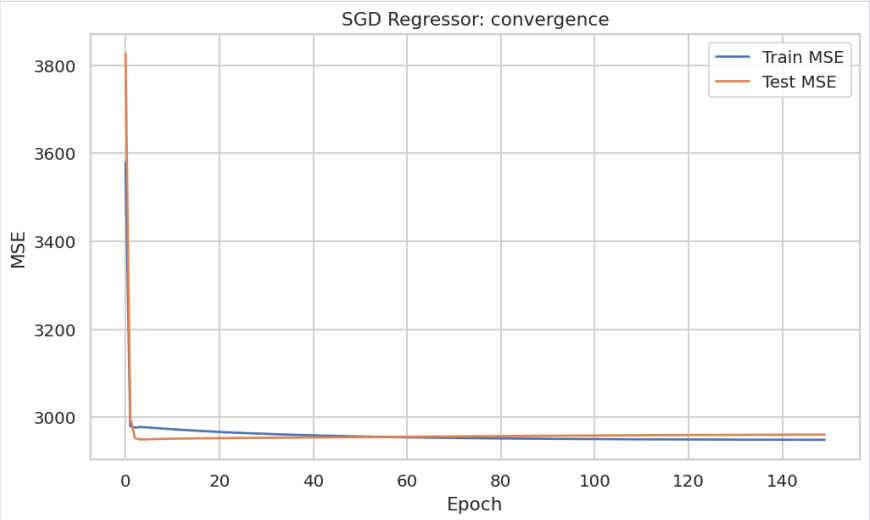


8) SGD Regressor

Loss per epoch

RMSE	54.41
MAE	43.28
R^2	0.441
Fit time (s)	—

Train and test MSE drop steeply in the first few epochs and then flatten near 2,950. The two curves stay close together, so there is no sign of overfitting, and the result is close to the other linear models.

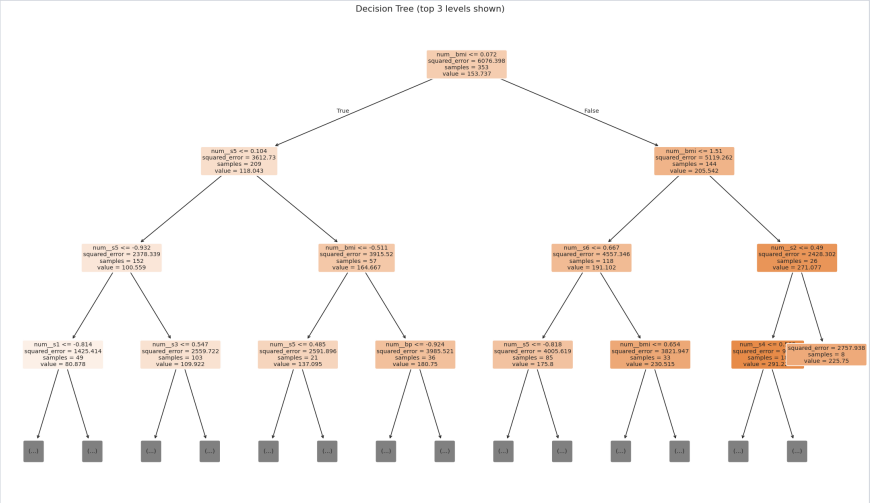


9) Decision Tree

The tree itself (top 3 levels)

RMSE	57.99
MAE	46.77
R^2	0.365
Fit time (s)	0.003

The root split is on BMI (at or below 0.072), followed by s5, s6 and BMI again. Colour darkens as the predicted value rises. A single tree is easy to read but has high variance, which explains its weaker R^2 of 0.365.

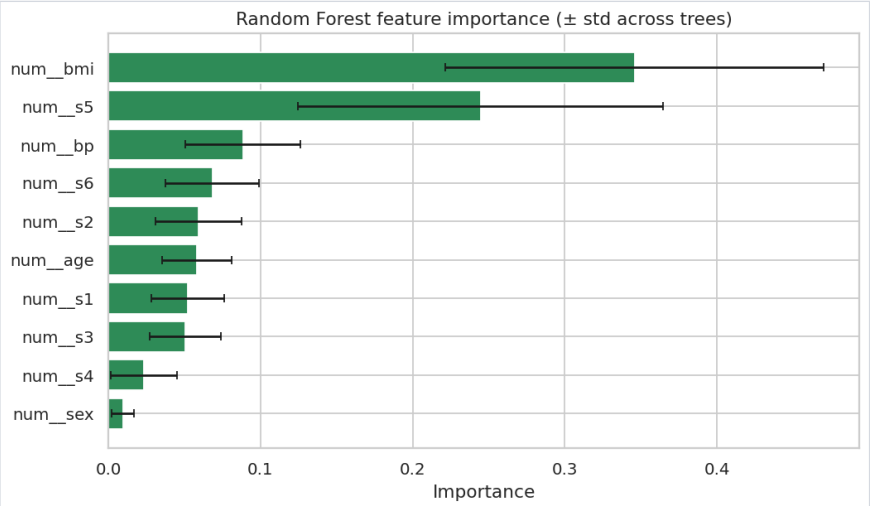


10) Random Forest

Feature importance with spread across trees

RMSE	54.92
MAE	44.75
R^2	0.431
Fit time (s)	0.641

BMI (~0.35) and s5 (~0.24) dominate; sex contributes least. The error bars (+/- std across trees) are large for the top two features, so their exact ranking varies between trees.

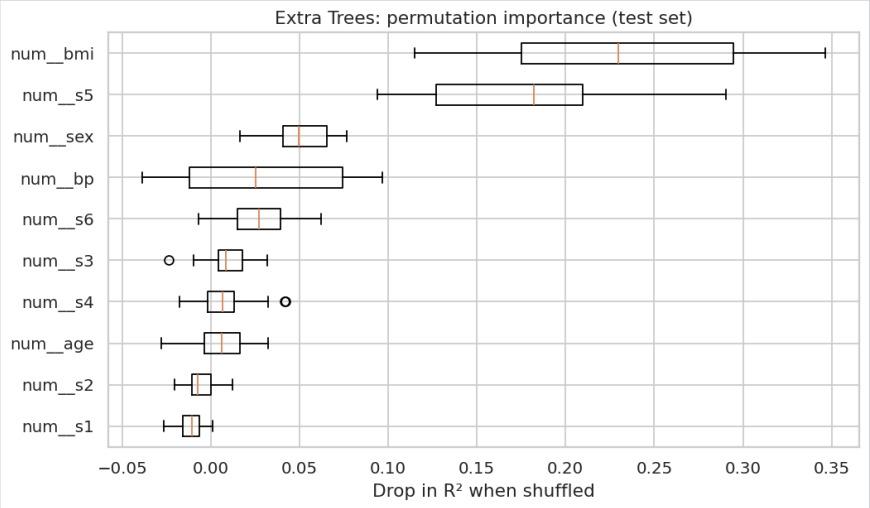


11) Extra Trees

Permutation importance boxplot

RMSE	53.05
MAE	43.86
R^2	0.469
Fit time (s)	0.386

Drop in test R^2 when each feature is shuffled. BMI (median ~0.23) and s5 (~0.18) matter far more than the rest. s1 and s2 sit at or just below zero, meaning they add nothing on the test set.

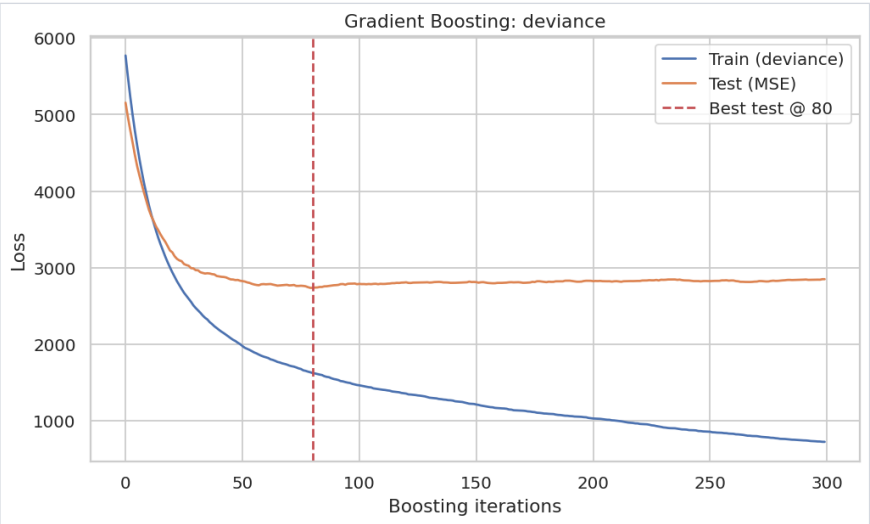


12) Gradient Boosting

Train vs test deviance by stage

RMSE	53.39
MAE	43.62
R^2	0.462
Fit time (s)	0.361

Training loss keeps falling over 300 iterations, while test MSE bottoms out at about iteration 80 and then stays flat. The growing gap shows overfitting, so fewer trees or early stopping would suffice.



13) Hist Gradient Boosting

Test R^2 by iteration

RMSE	57.87
MAE	46.09
R^2	0.368
Fit time (s)	0.233

Test R^2 peaks at 0.455 after roughly 60 iterations, then declines steadily to 0.368 at 300. The final score is held back by running too many iterations on 353 rows.

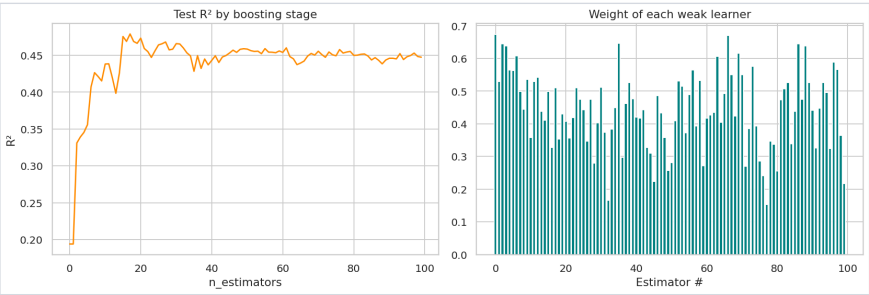


14) AdaBoost

Staged R^2 and per-estimator weights

RMSE	54.11
MAE	43.76
R^2	0.447
Fit time (s)	0.210

Left: test R^2 climbs quickly to about 0.45 within ~20 estimators and then plateaus. Right: the weight of each weak learner varies from about 0.15 to 0.7, so later learners still contribute unevenly.

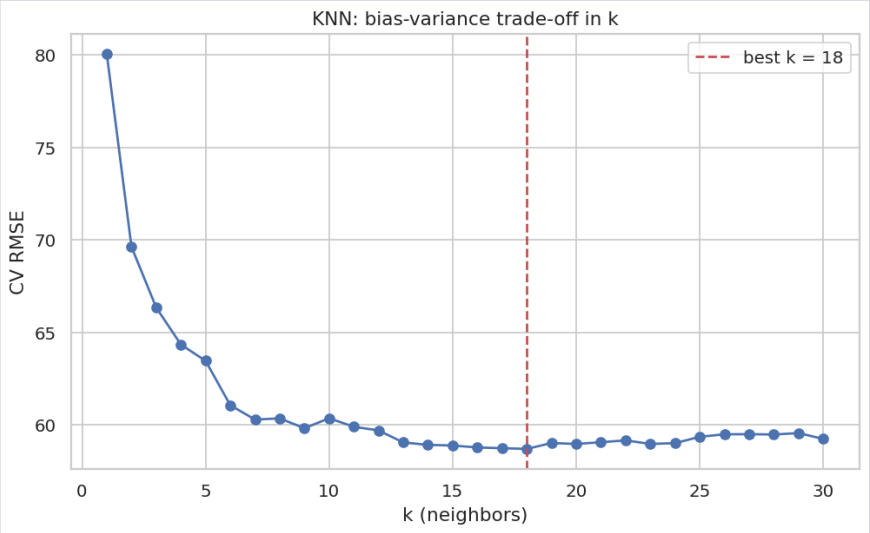


15) K-Nearest Neighbors

CV error vs k

RMSE	55.30
MAE	45.66
R^2	0.423
Fit time (s)	0.002

Classic bias-variance curve. k = 1 is very noisy (CV RMSE ~80), error falls as k grows and bottoms out at k = 18 (dashed line), then stays almost flat.

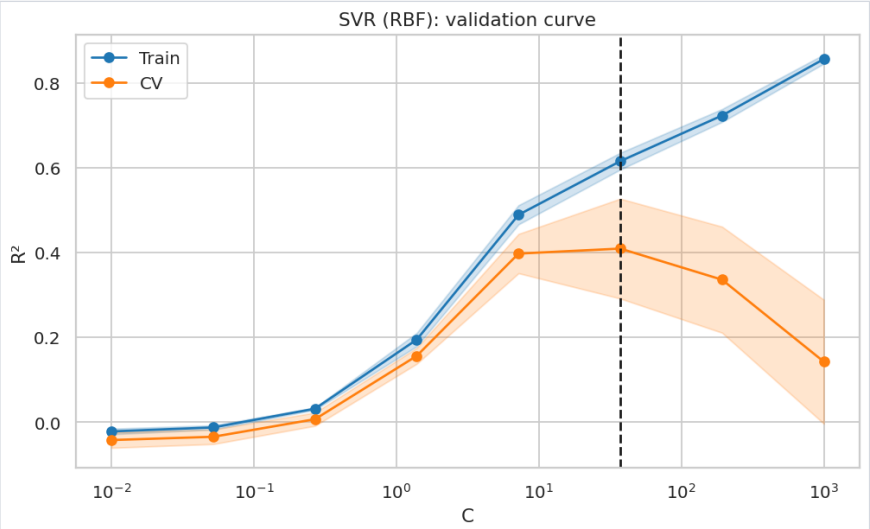


16) Support Vector Regression

Validation curve over C

RMSE	50.10
MAE	39.10
R^2	0.526
Fit time (s)	0.009

Train R^2 keeps rising with C, while CV R^2 peaks at about C = 40 (dashed line) and then drops, the signature of overfitting at large C. With tuned C this is the best model in the comparison on every metric.

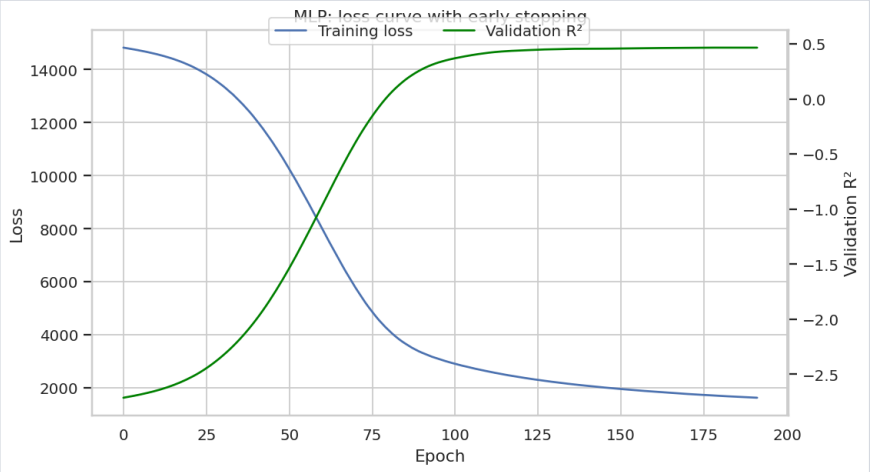


17) MLP Neural Network

Training loss & validation score

RMSE	60.47
MAE	48.16
R^2	0.310
Fit time (s)	0.303

Training loss falls smoothly and validation R^2 rises from strongly negative to a plateau, where early stopping applies. The weak test R^2 of 0.310 reflects the small training set and limited tuning.

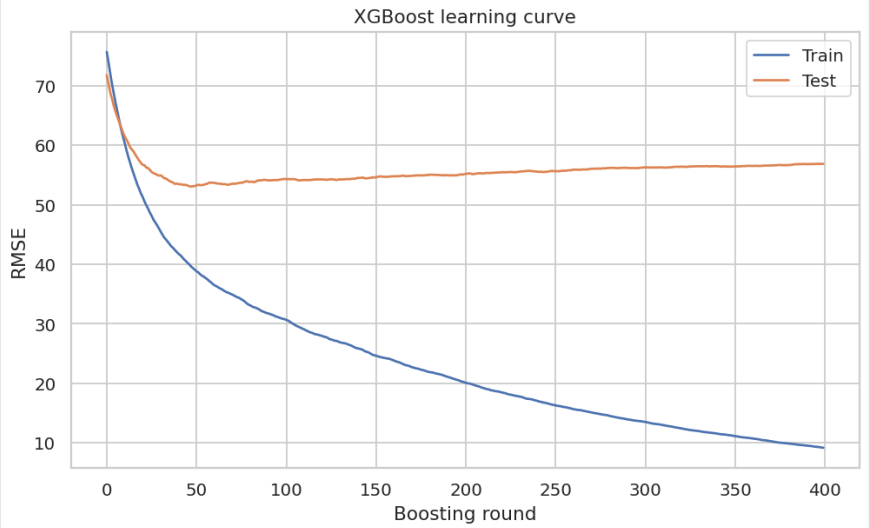


18) XGBoost

Learning curve (RMSE per round)

RMSE	56.85
MAE	45.49
R^2	0.390
Fit time (s)	0.150

Train RMSE drops to about 9 by round 400, while test RMSE bottoms near 53 around round 50 and creeps up to about 57. This is clear overfitting, and early stopping would likely recover a better score.

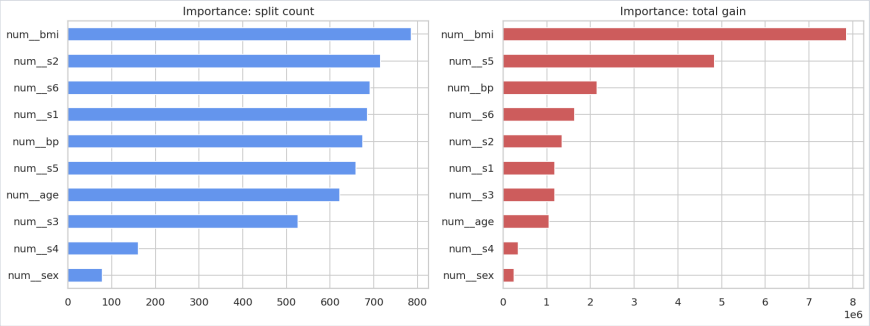


19) LightGBM

Feature importance by split count vs gain

RMSE	58.09
MAE	47.75
R^2	0.363
Fit time (s)	0.088

BMI ranks first by both split count and gain. By gain, s5 and bp are far more valuable than their split counts suggest, while s2, s6 and s1 are used often but add little gain.

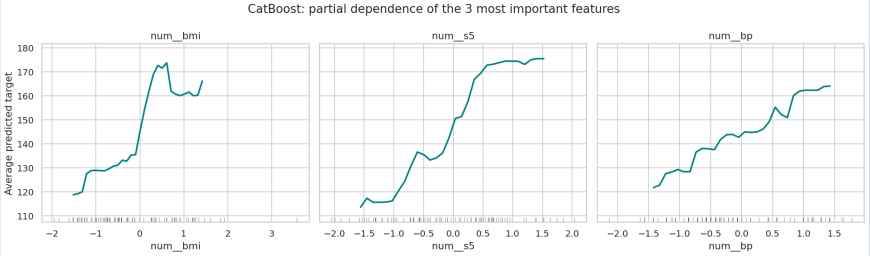


20) CatBoost

Partial dependence of the top features

RMSE	52.58
MAE	42.61
R^2	0.478
Fit time (s)	0.495

Predicted progression rises with BMI (a sharp jump around 0 to 0.5), with s5 (steep rise then plateau) and with blood pressure (steady, near-linear increase). These are the three most important features. Second-best model overall.

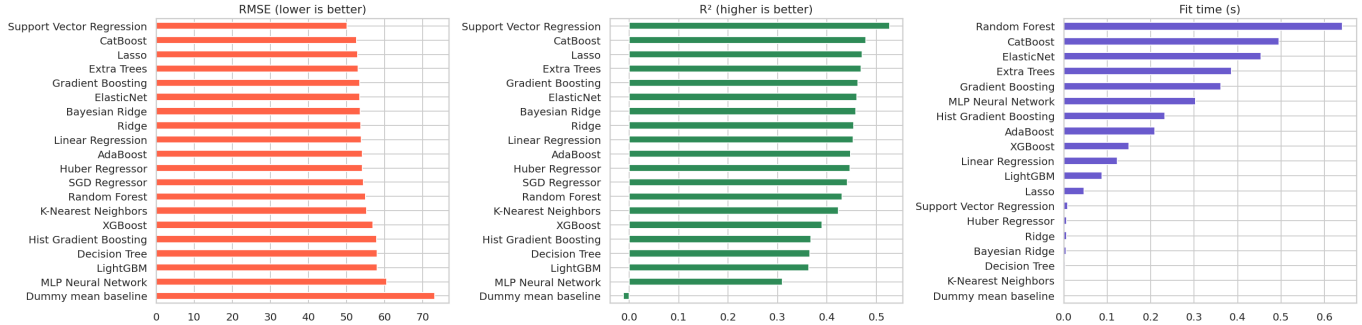


Model comparison dashboard

All 20 models ranked by test RMSE (lower is better)

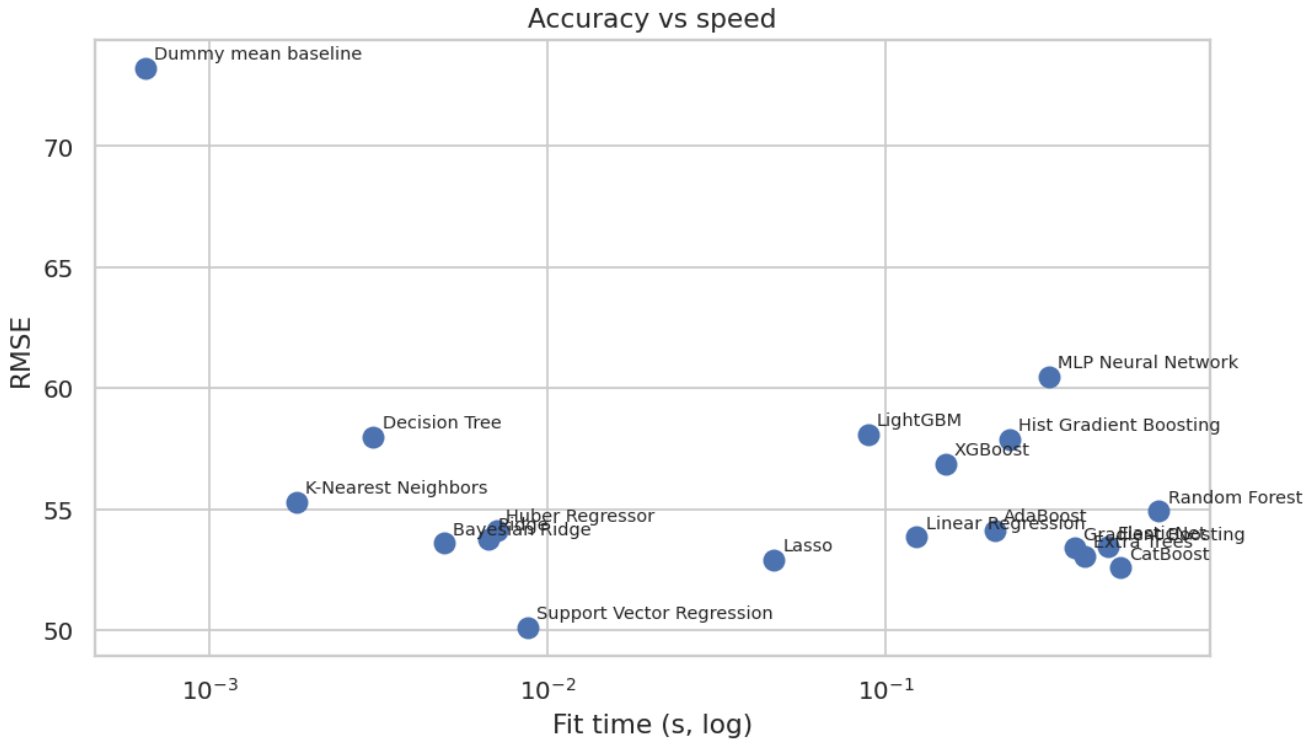
#	Model	RMSE	MAE	R^2	Fit (s)
1	Support Vector Regression	50.10	39.10	0.526	0.009
2	CatBoost	52.58	42.61	0.478	0.495
3	Lasso	52.92	42.79	0.471	0.047
4	Extra Trees	53.05	43.86	0.469	0.386
5	Gradient Boosting	53.39	43.62	0.462	0.361
6	ElasticNet	53.45	42.98	0.461	0.454
7	Bayesian Ridge	53.59	42.87	0.458	0.005
8	Ridge	53.77	42.81	0.454	0.007
9	Linear Regression	53.85	42.79	0.453	0.123
10	AdaBoost	54.11	43.76	0.447	0.210
11	Huber Regressor	54.14	42.88	0.447	0.007
12	SGD Regressor	54.41	43.28	0.441	—
13	Random Forest	54.92	44.75	0.431	0.641
14	K-Nearest Neighbors	55.30	45.66	0.423	0.002
15	XGBoost	56.85	45.49	0.390	0.150
16	Hist Gradient Boosting	57.87	46.09	0.368	0.233
17	Decision Tree	57.99	46.77	0.365	0.003
18	LightGBM	58.09	47.75	0.363	0.088
19	MLP Neural Network	60.47	48.16	0.310	0.303
20	Dummy mean baseline	73.22	64.01	-0.012	—

Key takeaways. Support Vector Regression (RBF) is best on this small dataset (RMSE 50.10, R^2 0.526), followed by CatBoost (52.58) and Lasso (52.92). Regularised linear models perform on par with tree ensembles, which suggests the signal is largely linear and the sample is small. The dummy baseline (R^2 about 0) confirms every model learns real structure. The MLP, LightGBM, Hist Gradient Boosting and Decision Tree trail, consistent with overfitting or limited tuning on 353 training rows.



Accuracy vs speed

RMSE against fit time (log scale)



Actual vs predicted for all models

